

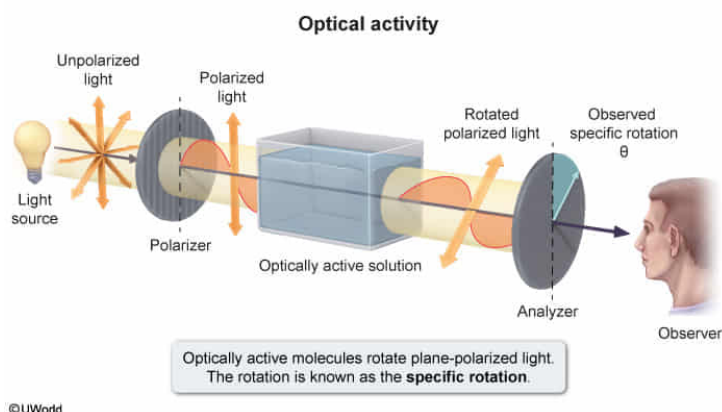
The experimental data obtained when a chiral molecule interacts with plane-polarized light is called:

- ☐ A. the polarimeter. [2%]
- ☐ B. the *R* or *S* configuration. [9%]
- ✓ ☒ C. the specific rotation. [85%]
- ☐ D. the infrared spectrum. [1%]

Correct

85% answered correctly

Explanation:



Normal light is made of light waves that oscillate in all planes perpendicular to the direction of propagation. When light encounters a polarizer, only waves that are aligned with the polarizer can pass. Therefore, all the light that passes through the polarizer oscillates in one direction. This resulting uniform light is known as **plane-polarized (linear) light**.

Chiral molecules rotate plane-polarized light; the angle of rotation, known as the **specific rotation**, is unique for each chiral molecule and must be determined **experimentally**. A molecule's specific rotation contains **direction** (+ or -) and **magnitude**. Rotations that are clockwise are (+) whereas those that are counterclockwise are (-). Enantiomers have specific rotations of equal magnitude but in opposite directions.

(Choice A) A polarimeter is the instrument used to measure the rotation of polarized light.

(Choice B) The *R* or *S* configurations are based on the priority ranking of substituents on a stereocenter, and are not correlated to the direction chiral molecules rotate plane-polarized light. Therefore, stereocenter configuration cannot be used to predict specific rotation, or vice versa.

(Choice D) An infrared spectrum is obtained from irradiation of a molecule with infrared light, causing bond vibrations and stretching at different frequencies. The amount of light absorbed by the molecule is recorded and displayed as the percent transmittance versus wavenumber on the infrared spectrum.

Educational objective:

Specific rotation measures the direction (+ or –) and magnitude (angle) of rotation by which chiral molecules rotate plane-polarized light. This value is a physical characteristic of chiral molecules, is unique for each molecule, and must be determined experimentally.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402291

System:
Introduction to Organic Chemistry

Suppose that a 0.2 g/mL solution of (S)-ibuprofen in a 2 dm long cell rotates plane-polarized light +10°. What is the specific rotation for (S)-ibuprofen?

- ☐ A. -50° [5%]
☐ B. -25° [14%]
☒ C. +25° [61%]
☐ D. +50° [18%]

Correct

60% answered correctly

Explanation:

Specific rotation

Standardized rotation that takes into account concentration & sample cell length.

$$[\alpha] = \frac{\alpha_{\text{obs}}}{c \times l}$$

$[\alpha]$ = Specific rotation (degrees)
 α_{obs} = observed rotation (degrees)
 c = concentration ($\frac{\text{g}}{\text{mL}}$)
 l = path length (dm)

$$[\alpha] = \frac{\alpha_{\text{obs}}}{c \times l} = \frac{+10^\circ}{0.2 \frac{\text{g}}{\text{mL}} \times 2 \text{ dm}} = \frac{+10^\circ \times \text{mL}}{0.4 \text{ g} \times \text{dm}} = +25^\circ$$

Note: Specific rotation units are $\frac{\text{degrees} \times \text{mL}}{\text{g} \times \text{dm}}$ but are often expressed as degrees for simplicity.
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Chiral molecules are optically active because they rotate **plane-polarized light**. The angle of rotation is known as the **specific rotation** and contains both direction (+ or -) and magnitude (degrees). A rotation in the clockwise direction is classified as (+) whereas counterclockwise rotation is classified as (-).

Specific rotation is unique to each chiral molecule and must be determined experimentally. The **observed rotation** is **dependent** on the sample **concentration** and **sample cell length**; the specific rotation is standardized to show what the rotation would be if the pathlength were 1 decimeter (dm) and the concentration were 1 g/mL. This is calculated via the following equation:

$$[\alpha] = \frac{\alpha_{\text{obs}}}{c \times l}$$

where $[\alpha]$ is the specific rotation, α_{obs} is the observed rotation (degrees), c is the concentration (g/mL), and l is the sample cell length (dm). The units of specific rotation are (degrees · mL) / (g · dm) but are typically expressed as degrees for simplicity.

The observed rotation for a 0.2 g/mL sample of (S)-ibuprofen in a 2 dm long cell is +10°. The specific rotation is calculated as follows:

$$[\alpha] = \frac{\alpha_{\text{obs}}}{c \times l} = \frac{+10^\circ}{0.2 \frac{\text{g}}{\text{mL}} \times 2 \text{ dm}} = +25^\circ$$

(Choices A and D) A specific rotation of -50° or +50° is obtained if the sample cell length is not accounted for and the observed rotation is divided only by the concentration. The specific rotation should be negative only when the observed rotation is negative.

(Choice B) A specific rotation of -25° is obtained by correctly using the specific rotation equation and dividing the observed rotation by the sample's concentration and sample cell length but then incorrectly changing the sign of the angle. Because the observed rotation is positive, the specific rotation should also be positive.

Educational objective:

Specific rotation is the angle by which chiral molecules rotate plane-polarized light and contains direction (+ or -) and magnitude (degrees). The observed rotation α_{obs} is dependent on the concentration (c in g/mL) and sample cell length (l in dm). Therefore, the specific rotation is standardized to account for these experimental parameters:

$$[\alpha] = \frac{\alpha_{\text{obs}}}{c \times l}$$

Time spent: 0 sec

QID:402292